

```
In [1]: import pandas as pd
import numpy as np

# NLP
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

# Machine Learning
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report

# Download NLTK data
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\Chint\AppData\Roaming\nltk_data...
[nltk_data] Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\Chint\AppData\Roaming\nltk_data...
[nltk_data] Unzipping corpora\stopwords.zip.
```

Out[1]: True

In []:

```
In [2]: data = {
    "ticket": [
        "Internet is not working",
        "Need refund for my payment",
        "App crashes while login",
        "Where is my order",
        "Password reset issue",
        "Unable to make payment",
        "Delivery is delayed",
        "System running very slow"
    ],
    "category": [
        "Technical",
        "Billing",
        "Technical",
        "General",
        "Technical",
        "Billing",
        "General",
        "Technical"
    ]
}

import pandas as pd
```

```
df = pd.DataFrame(data)

print(df)
```

	ticket	category
0	Internet is not working	Technical
1	Need refund for my payment	Billing
2	App crashes while login	Technical
3	Where is my order	General
4	Password reset issue	Technical
5	Unable to make payment	Billing
6	Delivery is delayed	General
7	System running very slow	Technical

```
In [3]: def assign_priority(text):
        text = text.lower()

        if "crash" in text or "not working" in text:
            return "High"

        elif "slow" in text or "delayed" in text:
            return "Medium"

        else:
            return "Low"

df["priority"] = df["ticket"].apply(assign_priority)

print(df)
```

	ticket	category	priority
0	Internet is not working	Technical	High
1	Need refund for my payment	Billing	Low
2	App crashes while login	Technical	High
3	Where is my order	General	Low
4	Password reset issue	Technical	Low
5	Unable to make payment	Billing	Low
6	Delivery is delayed	General	Medium
7	System running very slow	Technical	Medium

```
In [5]: import nltk

nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to
[nltk_data] C:\Users\Chint\AppData\Roaming\nltk_data...
[nltk_data] Unzipping tokenizers\punkt_tab.zip.
```

```
Out[5]: True
```

```
In [6]: from nltk.corpus import stopwords
        from nltk.tokenize import word_tokenize

        stop_words = set(stopwords.words('english'))

        def clean_text(text):
```

```

text = text.lower()

words = word_tokenize(text)

filtered_words = [
    word for word in words
    if word.isalnum() and word not in stop_words
]

return " ".join(filtered_words)

df["clean_ticket"] = df["ticket"].apply(clean_text)

print(df[["ticket", "clean_ticket"]])

```

	ticket	clean_ticket
0	Internet is not working	internet working
1	Need refund for my payment	need refund payment
2	App crashes while login	app crashes login
3	Where is my order	order
4	Password reset issue	password reset issue
5	Unable to make payment	unable make payment
6	Delivery is delayed	delivery delayed
7	System running very slow	system running slow

In [7]: `from sklearn.feature_extraction.text import TfidfVectorizer`

```

vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(df["clean_ticket"])

y = df["category"]

print(X.shape)

```

(8, 19)

In [8]: `from sklearn.model_selection import train_test_split`

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)

```

Training Data Shape: (6, 19)

Testing Data Shape: (2, 19)

In [9]: `from sklearn.naive_bayes import MultinomialNB`

```

model = MultinomialNB()

model.fit(X_train, y_train)

```

```
print("Model Trained Successfully")
```

Model Trained Successfully

```
In [10]: y_pred = model.predict(X_test)

print("Predicted Categories:")
print(y_pred)
```

Predicted Categories:
['Technical' 'Technical']

```
In [11]: from sklearn.metrics import accuracy_score, classification_report

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

Accuracy: 0.0

Classification Report:

	precision	recall	f1-score	support
Billing	0.00	0.00	0.00	2.0
Technical	0.00	0.00	0.00	0.0
accuracy			0.00	2.0
macro avg	0.00	0.00	0.00	2.0
weighted avg	0.00	0.00	0.00	2.0

```

C:\Users\Chint\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
    _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\Chint\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero_division` parameter to control this behavior.
    _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\Chint\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
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C:\Users\Chint\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
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    _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])

```

```

In [12]: new_ticket = ["Payment failed during recharge"]

# Clean text
cleaned = [clean_text(text) for text in new_ticket]

# Convert to TF-IDF vector
vector = vectorizer.transform(cleaned)

# Predict category
prediction = model.predict(vector)

print("Predicted Category:", prediction[0])

# Priority
priority = assign_priority(new_ticket[0])

print("Priority:", priority)

```

Predicted Category: Technical
Priority: Low

In []: